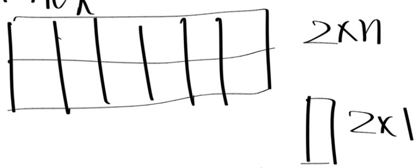
生成函数

回顾：



f_n = # ways to cover 2xn
with 2x1 Domino

f_0 1

f_1 ▯

f_2 ▯▯ ⊟ f_3 ▯▯▯ ▯⊟ ⊟▯

$f_n = f_{n-1} + f_{n-2}$

1 + ▯ + ▯▯ + ⊟ + ▯⊟ + ⊟▯ ⋯ $= 1+x+x^2+y^2+xy^2+y^2x+\cdots = \sum_{m} (???)\, x^n y^{2m}$

= 1 + (▯ + ▯⊟ + ▯▯ + ▯▯▯ ⋯) + (⊟, ⊟▯ ⋯)   只是一个形式幂级数

= 1 + ▯F + ⊟F $= 1+xF+y^2F$

F = 1 / (1 − ▯ − ⊟) $= \frac{1}{1-x-y^2}$

▯ $=x$, ▭ $=y$, ⊟ $=y^2$

▯ $=x$, ▭ $=x$, ⊟ $=x^2$

$$F = \frac{1}{1-x-x^2} = \sum_{n=0}^{\infty} f_n x^n$$

Generative functions

$a_0, a_1, a_2, \cdots$

$A(x) = \sum_n a_n x^n.$

e.g. Seq GF

$1,0,0,\cdots$ $A(x)=1$

$\underbrace{0,\cdots,0}_{m-1},1,0,\cdots$ $A(x)=x^m$

$1,1,1,\cdots$ $A(x)=\frac{1}{1-x}$

$1,\alpha,\alpha^2,\alpha^3,\cdots$ $\frac{1}{1-\alpha x}$

$1,0,1,0,1,0\cdots$ $\frac{1}{1-x^2}$

$\binom{p}{0},\binom{p}{1},\binom{p}{2},\cdots$ $(1+x)^p$

$\binom{c}{0}\binom{c+1}{1}\cdots$ $\frac{1}{(1-x)^{c+1}}$

Manipulation

$g_0, g_1, g_2, \cdots$ $\quad G(x)$

$0, g_0, g_1, \cdots$ $\quad xG(x)$

$g_1, \cdots$ $\quad \frac{G(x)-g_0}{x}$

$\cdots \alpha g_n+\beta h_n, \cdots$ $\quad \alpha G(x)+\beta H(x)$

$g_0, g_1\alpha, g_2\alpha^2 \cdots$ $\quad G(\alpha x)$

$g_0, 0, g_1, 0, g_2 \cdots$ $\quad G(x^2)$

$G(x)=g_0+g_1x+\cdots$

$[x^n]G(x)=g_n$.

$[x^n]G(x)H(x)$

$=g_0h_0+(g_0h_1+g_1h_0)x+\cdots$

$=\sum_t g_t h_{n-t} x^n$

找零问题：trivial的例子

①①①① $\quad 1+x+x^2+x^3+x^4$

⑤×∞ $\quad 1+x^5+x^{10}+\cdots=\frac{1}{1-x^5}$

$$\frac{1+x+x^2+x^3+x^4}{1-x^5}=\frac{1}{1-x}$$

处理方式，拆成 $\frac{\text{常数}}{\text{一次项}}$ (有理式).

如 $\frac{1}{1-x-x^2}$ 这种 $\quad$ ↳ 然后 ez.

1,0,1,0 ⋯ 还可看成 1,1,⋯

1,-1,1,-1 ⋯

平均 ⇒ $\frac{1}{2}\left(\frac{1}{1-x}+\frac{1}{1+x}\right)$

补 Catalan, $C_n=\sum_{i+j=n-1} C_iC_j$

$C_0=1$

$$C(x)=\sum_{n=0}^{\infty} C_nx^n=1+\sum_{n=1}^{\infty}\sum_{i=0}^{n-1} C_iC_{n-1-i}x^n$$

$$=1+x\sum_{n=0}^{\infty}\sum_{i=0}^{n} C_iC_{n-i}x^n$$

$$=1+xC(x)C(x)$$

$$xC^2(x)-C(x)+1=0$$

$C(x)=\frac{1-\sqrt{1-4x}}{2x}$ $\quad$ 可用广义二项式展开

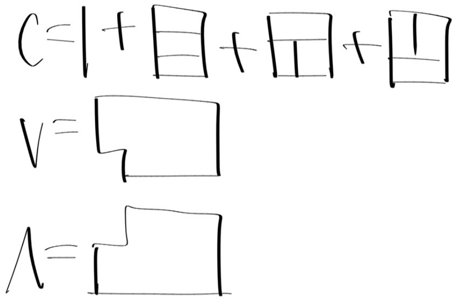
Manipulation补充:

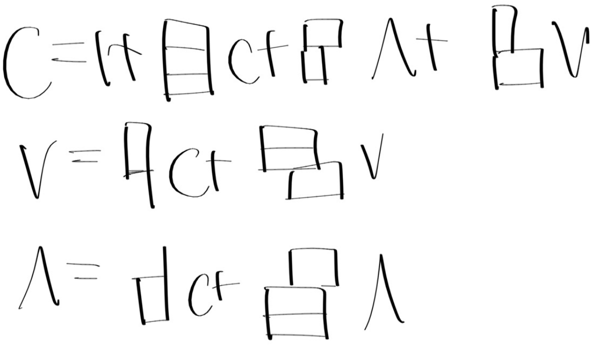
$\sum g_i \cdots \quad \frac{1}{1-x} G(x)$

$\cdots (n+1) g_{n+1}, \cdots \quad G'(x)$

$\cdots, n g_n, \cdots \quad x G'(x)$

3xn.





$C(x) = 1 + x^3 C(x) + x^2 V(x) + x^2 \Lambda(x)$

$V(x) = x C(x) + x^3 V(x)$

$V(x) = \frac{x}{1-x^3} C(x)$

$C(x) = 1 + x^3 C(x) + 2x \frac{x}{1-x^3} C(x)$

$g_0, g_1, g_2, \cdots \longleftrightarrow \overset{GF}{\sum_n} g_n x^n$

exponential GF

$\hat{G}(x) = \sum_{n=0}^{\infty} g_n \frac{x^n}{n!}$

$n! [x^n] \hat{G}(x) \hat{H}(x)$

$$\overset{\|}{=} \sum_{i=0}^{n} \frac{g_i}{i!} \frac{h_{n-i}}{(n-i)!} n!$$

$$= \sum_{i=0}^{n} \binom{n}{i} g_i h_{n-i}$$

$1 + \hat{G}(x) + \frac{\hat{G}^2(x)}{2!} + \cdots$

$\|$

$e^{\hat{G}(x)}$

$\overbrace{1+1+\cdots+1}^{n} = n \quad 0! B_0 \frac{n}{1!0!}$

$1+2+\cdots+n = \frac{n(n+1)}{2} \quad 1!\left(B_0 \frac{n^2}{2!0!} + B_1 \frac{n}{1!1!}\right)$

$1^2+2^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6} \quad 2!\left(B_0 \frac{n^3}{3!0!} + B_1 \frac{n^2}{2!1!} + B_2 \frac{n}{1!2!}\right)$

$1^3+2^3+\cdots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$

$$\sum_{m=0}^{\infty} \sum_{0 \le k < n} k^m \frac{x^m}{m!}$$

$$= \sum_{0 \le k < n} \sum_{m} k^m \frac{x^m}{m!} \quad e^{kx}$$

$$= \sum_{0 \le k < n} e^{kx} = \frac{1-e^{xn}}{1-e^x}$$

$$= \left(\frac{x}{e^x-1} \cdot \frac{e^{xn}-1}{x}\right)$$

$$e^{xn}-1 = n\frac{x}{1} + n^2 \frac{x^2}{2!} + n^3 \frac{x^3}{3!} + \cdots$$

$$\frac{x}{e^x-1} = B_0 \cdot \frac{1}{0!} + B_1 \frac{x}{1!} + B_2 \frac{x^2}{2!} + \cdots$$

$$\sum_{0 \le k < n} k^m = \cdots\cdots$$